

ZCU216 RFSoc

tProcessor V2
Pulse sequencing

Signal gen 0
DAC tile 2, blk 0

Signal gen N
Qubit drive, π pulse

IQ output
Decimated/accum.

Readout ch. 0
ADC tile 2, blk 0

XM655

BALUN (RO TX)
diff. $\rightarrow$ single

BALUN (DR TX)
diff. $\rightarrow$ single

BALUN (RX)
single $\rightarrow$ diff.

Loopback (cal.):
TX $\rightarrow$ SMA $\rightarrow$ RX

Qubit chip (future)

Resonator
Dispersive readout

Qubit
Fluxonium/transmon

g

 Readout drive

 Qubit drive

 Readout return

 Dispersive coupling